

ERD Explanation point by point

- **1 Define the scope of your societal problem**

The goal of our database is to investigate how rising energy prices, inflation, and central-bank policies relate to the financial pressures faced by UK households. The constructed database will allow us to compare household energy spending and recorded borrowing costs across regions and demographic groups. Some of the concrete questions we will attempt to answer are as follows: How do household energy bills change over time and compare among regions? How does energy consumption and spending differ between vulnerable households?

The database supports comparisons, but cannot establish cause and effect by itself or calculate exact energy spending because it stores income brackets rather than exact incomes of households.

- **2 List how the data interacts in short sentences**

- A region can have many macroeconomic records, but each record belongs to one region.
- A region can contain many households, but each household belongs to one region.
- A household can receive many energy bills, but each bill belongs to one household.
- An energy supplier can issue many bills, but each bill references one supplier.
- A household can have several demographic profiles, and a profile can describe several households.
- A commodity shock can be linked to several macroeconomic records, and a macroeconomic record can be linked to several shocks.
- A central-bank policy can have recorded impacts on several households, and a household can have recorded impacts from several policies.

- **3 Identify the entities**

The eight core entities are REGION, HOUSEHOLD, DEMOGRAPHIC_PROFILE, ENERGY_BILL, ENERGY_SUPPLIER, MACROECONOMIC_METRIC, COMMODITY_SHOCK, and CENTRAL_BANK_POLICY. Three bridge entities are HOUSEHOLD_DEMOGRAPHIC, SHOCK_IMPACT_LOG, and HOUSEHOLD_POLICY_IMPACT.

- **4 Identify the relationships and their cardinalities**

Relationships

Entities	Relationship	Cardinality
Region to Household	Contains	One-to-many
Region to Macroeconomic metric	Has measurements	One-to-many
Household to Energy bill	Receives	One-to-many
Energy supplier to Energy bill	Decides	One-to-many
Household and Demographic profile	Is classified by	Many-to-many
Commodity shock to Macroeconomic metric	Has an impact on	Many-to-many

Entities	Relationship	Cardinality
Central-bank policy and Household	Has an impact on	Many-to-many

Each child record references exactly one parent through its required FK. A parent can exist without a related child. For example, a household can be recorded before any energy bills are added.

- **5 Bridge tables**

- HOUSEHOLD_DEMOGRAPHIC connects households to demographic profiles and records the assignment date.
- SHOCK_IMPACT_LOG connects commodity shocks to macroeconomic records and stores the impact type and value.
- HOUSEHOLD_POLICY_IMPACT connects policies to households and stores borrowing costs and mortgage rates.

Each many-to-many relationship becomes two one-to-many relationships that produces a total of 10 relationships shown in the ERD.

- **6 Decide which attributes to include per entity**

We need to account for circumstances of households, energy consumptions and spendings, economic conditions and recorded impacts. Billing periods and recording dates allow comparisons across time to draw comparisons.

Entity	Attributes
REGION	region_id, region_name, country, total_population
HOUSEHOLD	household_id, region_id, postcode, dwelling_type, occupant_count, housing_tenure
DEMOGRAPHIC_PROFILE	demographic_id, income_bracket, age_group, vulnerability_flag
HOUSEHOLD_DEMOGRAPHIC	household_id, demographic_id, assigned_date
ENERGY_SUPPLIER	supplier_id, supplier_name, ofgem_license_code, support_scheme_enrolled
ENERGY_BILL	bill_id, household_id, supplier_id, billing_period_start, billing_period_end, kwh_consumed, price_cap_rate, total_amount_gbp
MACROECONOMIC_METRIC	metric_id, region_id, recording_date, headline_cpi, inflation_rate, energy_cpi_weight
COMMODITY_SHOCK	shock_id, event_name, commodity_type, shock_start_date, shock_end_date, wholesale_price_index
SHOCK_IMPACT_LOG	shock_id, metric_id, impact_type, impact_value
CENTRAL_BANK_POLICY	policy_id, effective_date, base_interest_rate, mortgage_stress_index
HOUSEHOLD_POLICY_IMPACT	policy_id, household_id, borrowing_cost_gbp, mortgage_rate

- **7 Primary keys and Foreign keys**

Every table has a unique primary key. Each composite bridge key prevents duplicate records. ofgem_license_code is an alternate key and must be unique.

- **8 ERD in Crow foot notation**

Primary and foreign keys

Entity	Primary key	Foreign keys
REGION	region_id	None
HOUSEHOLD	household_id	region_id to REGION
DEMOGRAPHIC_PROFILE	demographic_id	None
HOUSEHOLD_DEMOGRAPHIC	household_id, demographic_id	household_id to HOUSEHOLD; demographic_id to DEMOGRAPHIC_PROFILE
ENERGY_SUPPLIER	supplier_id	None
ENERGY_BILL	bill_id	household_id to HOUSEHOLD; supplier_id to ENERGY_SUPPLIER
MACROECONOMIC_METRIC	metric_id	region_id to REGION
COMMODITY_SHOCK	shock_id	None
SHOCK_IMPACT_LOG	shock_id, metric_id	shock_id to COMMODITY_SHOCK; metric_id to MACROECONOMIC_METRIC
CENTRAL_BANK_POLICY	policy_id	None
HOUSEHOLD_POLICY_IMPACT	policy_id, household_id	policy_id to CENTRAL_BANK_POLICY; household_id to HOUSEHOLD

- **9 Supertypes and subtypes**

Supertypes were decided to not be necessary as the entities describe distinct objects and there is no clear shared parent that would make the design easier.

- **10 Normalize the ERD**

- **1NF form requires atomic values**

Each field should contain one value. Multiple demographic memberships must be stored as separate records.

Before

household_id PK	postcode	demographic_ids violation
101	NE1 1AA	201, 202
102	NE1 2BB	201

Violation: household 101 has two demographic IDs in one cell. Household 101 cannot act as a single demographic foreign key, and queries would need to separate the list to locate each membership.

After

household_id PK part	demographic_id PK	postcode
101	201	NE1 1AA
101	202	NE1 1AA
102	201	NE1 2BB

The fields are now atomic. The pair of household_id and demographic_id identifies each membership. However, the household postcode is repeated, creating the partial dependency.

- 2NF removes partial dependencies

After 1NF, non-key attributes must depend on the complete key. With a composite key of household_id, demographic_id, household or profile facts that depend on only one ID belong in their parent table.

Before

household_id PK	demographic_id PK	postcode violation	income_bracket violation	assigned_date
101	201	NE1 1AA	Low	2026-01-01
101	202	NE1 1AA	Low	2026-01-01
102	201	NE1 2BB	Low	2026-02-01

violations: postcode depends only on household_id and income_bracket depends only on demographic_id.

After

household_id PK	postcode
101	NE1 1AA
102	NE1 2BB

After DEMOGRAPHIC_PROFILE

demographic_id PK	income_bracket	age_group	vulnerability_flag
201	Low	18–64	True
202	Low	65+	True

After HOUSEHOLD_DEMOGRAPHIC

household_id PK FK	demographic_id PK FK	assigned_date
101	201	2026-01-01
101	202	2026-01-01
102	201	2026-02-01

Household and profile information is now stored independently. assigned_date remains in the bridge entity because it describes a particular household–profile.

- 3NF removes transitive dependencies

After 2NF, transitive dependencies must be removed. Supplier details should depend on the supplier key and be maintained in the supplier table.

Before supplier info repeated on bills

bill_id PK	household_id	supplier_id	supplier_name violation
401	101	301	North Energy
402	101	301	North Energy
403	102	301	North Energy

violation: bill_id determines supplier_id, which determines supplier_name. Updating a supplier's name would require changing multiple bill records.

After ENERGY_SUPPLIER

supplier_id PK	supplier_name
301	North Energy

After ENERGY_BILL

bill_id PK	household_id FK	supplier_id FK
401	101	301
402	101	301
403	102	301

Supplier information is stored once, while bills retain the foreign key. This avoids inconsistent supplier names across bills and allows a supplier record to exist even before it has any bills. The same principle supports separating regional details into REGION and shared policy details into CENTRAL_BANK_POLICY.

• 11 ERD still meets the scope requirements

The ERD supports the comparisons through its household, demographic, billing, regional, shock, and policy records. All entities remain connected.